SE311 SPRING 2025-2026 / 16-04-2026
Lab Work 07 – Adapter Pattern

You are developing a **FlightControlSystem** for a new jet. This software is designed to work with any sensor that follows the **SpeedSensor** interface. This interface is modern and expects a method called **recordVelocity(double kmh)**, where the speed is provided as a decimal in kilometers per hour. The entire dashboard is calibrated to display these metric values.

However, the new jet is using a high-precision old sensor called **ImperialSensor** (a speed sensor) salvaged from an older aircraft. This legacy system does not understand kilometers per hour. Instead, it uses a method called **logSpeedKnots(int knots, String pressureReading)**. It requires the speed to be a whole number in knots (nautical miles per hour) and demands an atmospheric pressure reading as a string, which the modern system doesn't even track.

Because the **FlightControlApp** only knows how to send metric decimals to a **recordVelocity** method, it is impossible for the dashboard to get data from the imperial sensor. The code is broken because the method names don't match, the units are different (KMH vs. Knots), and the data types are incompatible (Double vs. Integer/String).

Your job is to use **class Adapter** and try to adapt the **ImperialSensor** class into the **FlightControlSystem**.

Finally, test your code in your main with creating objects from **FlightControlSystem** and other **SpeedSensor** objects.